



Re-Layout Finished Goods Warehouse Using Dedicated Storage and Class-Based Storage Method (Study Case at PT. Sari Warna Asli V Textile Industry)

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The problems that occur in PT Sari Warna Asli V Textile Industry are goods that are scattered and not properly arranged, resulting in officers having difficulty finding goods and disrupting the smoothness and efficiency of operational activities, as well as the slow release of goods. For this reason, this research is conducted to improve the layout of the finished goods warehouse using dedicated storage and class-based storage method, and to compare the old layout and the two proposed layouts to find the layout that provides the most optimal results to be applied in the finished goods warehouse of PT Sari Warna Asli V Textile Industry. The method used to solve the problem in this study is a comparative quantitative method. The result of this research is the proposed layout with dedicated storage method results in a travel distance of 40.4 m. While the proposed layout with class-based storage method produces a distance of 14.84 m. Therefore, it can be concluded that the layout with class-based storage method is suitable to be applied in the finished goods warehouse of PT Sari Warna Asli V Textile Industry.

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1. INTRODUCTION

The warehouse has a function as a storage location for supplies that are critical to a company before they are distributed to their destination (Juliana, 2016). In order to maintain the quality and safety of the stored goods, warehouses must be properly managed according to the basic warehouse management and established guidelines. One of the things

that needs to be considered in warehouse management is the layout in the warehouse. The placement of goods in the warehouse cannot be done carelessly because each item has different characteristics (Shobah, 2021). In addition, a good warehouse layout can optimize the use of space, time, human resources and equipment (Aulia et al., 2023).

PT Sari Warna Asli V Textile Industry is a textile company engaged in thread spinning. The company has three warehouses, namely raw material warehouse, finished goods warehouse and spare parts warehouse. There are problems in the finished goods warehouse where the goods are not organized properly and are disorganized, which makes it difficult for officers to find goods and disrupts the smoothness and efficiency of operational activities, as well as the slow release of goods. According to Basuki & Hudori (2016), other losses that occur include the possibility of damage to goods, as well as suboptimal movement time and cost. In addition, according to Hadiguna & Setiawan (2008), paying attention to the layout of the warehouse is one way to achieve the effectiveness and efficiency of a business unit.

There are several methods of determining the warehouse layout, including dedicated storage (DS) and class-based storage (CBS). In the dedicated storage method, inventory is stored in a fixed location and each item has its own storage area based on the activity of the item (Husin, 2020 & Harianja, 2021). However, the space cannot be used optimally because the storage location of the goods cannot be changed even if there is free space (Ariyanto, 2023). Several studies have shown that the use of the dedicated storage method can reduce the total mileage by 28.08% and 50.72% (Nursyanti, 2019 & Ernawati, 2023). The application of dedicated storage methods can also reduce the search time for goods (Yusriski & Pardiyono, 2022), save storage area and expand the mobility area, and facilitate the retrieval of goods (Hidayat & Yanto, 2022).

Another method that can be used in determining the warehouse layout is class-based storage where this method classifies inventory into product classes based on appropriate criteria such as volume or usage rate (Bahrami, 2019 & Sitorus et al., 2020). Pareto's law can be used in the inventory classification process by dividing the inventory into three classes namely A, B and C (Harma, 2020). Items in group A or fast moving items can be stored near the shipping area while group B and C items are placed in the middle and back areas (Saidatuningtyas & Primadhani,

2021). According to McInerney, et al (2022), storing fast-moving items near the shipping area can reduce picking time by an average of 3 hours 20 minutes per day, thereby increasing goods movement (throughput) and warehouse profitability. Ariyanto (2023) explained that the use of class-based storage method can optimize warehouse utilization where there is a decrease in the need and cost of purchasing shelves in a drug storage warehouse. In addition, Isnaeni & Susanto (2021) mentioned that the application of class-based storage method provides savings in movement distance and makes it easier for officers to search and monitor goods. The results of the research conducted by Alfatiyah, et al (2022) also showed a reduction in travel distance by 29.58%.

Thus, based on previous studies, the definition of dedicated storage method is a method that focuses on organizing warehouses based on the number of activities in and out of goods with the shortest distance of movement. Meanwhile, the class-based storage method places goods based on the similarity of goods, such as material type, method of movement, and so on. Both methods have in common that product placement is fixed location. Referring to the above description, the researcher wants to conduct a study entitled: "Re-Layout of Finished Goods Warehouse with Dedicated Storage and Class-based Storage Methods (Case Study: PT. Sari Warna Asli V Textile Industry)". The researchers will compare two methods, namely dedicated storage and class-based storage methods, to find a warehouse layout that is suitable for implementation.

2. LITERATURE REVIEW

A warehouse is a place to store goods, while warehousing is an activity to manage goods in a warehouse that aims to maintain the quality and quantity of goods, maintain the orderliness of goods, improve distribution services, and provide accurate, actual, and accountable data and information. To perform its functions well, warehouses need to be managed appropriately, or commonly referred to as warehouse management. The scope of warehouse management includes (Muntasir, 2019):

- a. Human resource management.
 - 1) Effective utilization of human resources.

- 2) Minimizing the risk of occupational accidents.
- 3) Goods supervision and monitoring.
- b. Arrangement of goods, including receipt, storage and release of goods.
 - 1) Arrange the layout of goods to optimize the use of space.
 - 2) Minimize loss and damage to goods.
 - 3) Organize storage activities (throughout).

Storage policies for goods in warehouses are grouped into four categories according to Francis, et al (1992):

a. Randomized Storage

This policy is implemented by placing new items in empty spaces so that all spaces have an equal chance of being selected. This policy has a high utility of space utilization, but officers have difficulty finding the items they are looking for, thus slowing down the warehouse operations. The same thing was also conveyed by Anisa (2023), she stated that randomized storage method takes more time because the transfer of goods does not use certain standards. This method is suitable for organizations or companies with a small storage scale and is less suitable for organizations that have SOPs (Standard Operating Procedures).

b. Dedicated Storage

Dedicated storage is a storage policy commonly referred to as fixed storage because each item has its own storage area. Implementing this policy requires a large amount of storage space because empty space cannot be filled with different types of goods.

c. Class-Based Storage

This policy is a combination of randomized storage and dedicated storage policies based on the length of time goods are in the warehouse. The utility level of this policy is high and is suitable for companies that have limited space, but the entry and exit times of goods must be known with certainty. The main thing that must be taken into account is the difference in time of goods entry and exit (Anisa, 2023). For this reason, it is necessary to have a special officer who is always monitoring the time of entry and exit of goods.

d. Shared Storage

In class-based storage policy, goods and spaces are grouped into classes according to the similarity of the goods, such as type, displacement type, popularity, and others. The grouping of goods can be done using the Pareto law by dividing the goods into three groups (A, B, and C) based on the highest to lowest throughput values, which contain 80%, 15%, and 5%, respectively. This type of storage is more flexible because other goods can occupy the storage space of other goods, provided that the goods are of the same class.

3. RESEARCH METHOD

The location in this research is at PT Sari Warna Asli V Textile Industry. The object of this research is the layout of facilities and goods in the finished goods warehouse of polyester thread including all activities of storage and release of goods and material handling. The data used are primary data obtained directly from PT Sari Warna Asli V Textile Industry. The data includes:

- a. Overview of the current layout of the finished goods warehouse of PT Sari Warna Asli V Textile Industry
- b. Data on the types of goods stored
- c. Dimensions of stored goods and their storage capacity
- d. Dimensions of warehouse, racks and pallets
- e. Incoming, outgoing and fast moving goods data
- f. Material handling equipment used.

Steps in data processing are:

- a. Calculation of space requirements (S_j) and throughput (T_j)

This calculation is performed to determine the space requirements for each item stored and to determine the frequency of movement of each item.
- b. Dedicated storage method analysis

After calculating the space requirements and throughput, calculations are performed using the formula in the dedicated storage method and sorting from the largest to the smallest throughput value. In this activity, the researchers also calculated the distance of goods movement from the exit.

c. Analysis Using the Classified Storage Method

In this step, the throughput values are grouped into three classes (A, B, and C) and then sequentially placed in the area closest to the farthest from the exit. After that, the researcher calculates the distance of the goods movement from the exit.

d. Overview of Proposed Layout

After the analysis using the dedicated storage method and the class-based storage method, the proposed layout can be described in the form of a two-dimensional image using Microsoft Visio.

e. Comparison of the Current Layout and the Two Proposed Layouts

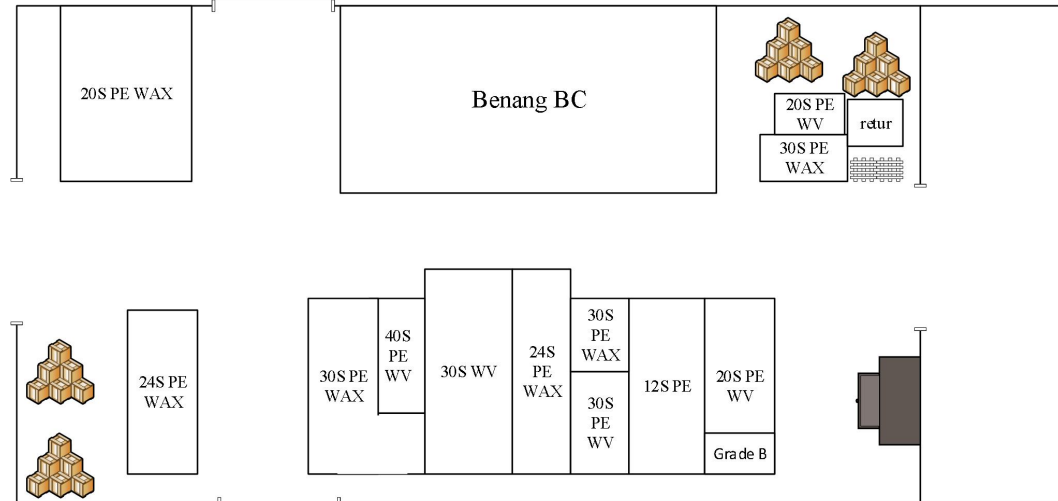
The last step is to compare the three existing layouts to get the best layout.

Table 1. Average daily inventory

Thread Type	Daily Average per Month (<i>bale</i>)			Total (<i>bale</i>)	Daily Average (<i>bale</i>)
	February	March	April		
GRADE BL	7,69	7,74	7,82	23,25	7,75
12S PE WV	6	6	6	18	6
20S PE WV	76,99	12,91	0,00	89,9	29,97
20S PE WAX	234,73	316,57	210,71	762,01	254
24S PE WAX	112,67	290,11	388,00	790,78	263,59
30S PE WV	22,79	43,62	125,61	192,02	64,01
30S PE WAX	121,61	93,81	459,81	675,23	225,08
40S PE WV	6,27	8,57	20,14	34,98	11,66
Total				2586,17	862,06

Source : Primer Data of PT. Sari Warna Asli Industry (data analyzed by the author)

Figure 1. Overview of The Current Layout of The Finished Goods Warehouse



Source : PT. Sari Warna Asli V Textile Industry

4. RESULT AND DISCUSSION

4.1 Calculation of Space Requirements and Throughput

Before calculating space requirements, it is necessary to know the width of the aisle in the warehouse. The aisle is used as the material handling path. The width of the aisle can be determined based on the size of the material handling equipment or hand lift used. PT. Sari

Warna Asli V Textile Industry uses hand lift with size 115 cm x 55 cm x 120 cm and calculated using the following formula:

$$\begin{aligned}
 \text{Aisle} &= \sqrt{\text{length}^2 + \text{width}^2} \dots \dots \dots (1) \\
 &= \sqrt{1,15^2 + 0,55^2} \\
 &= 1,452 \text{ m} \approx 1,5 \text{ m}
 \end{aligned}$$

Tompkins (2010) recommends a similar

aisle width for manual material handling of 5 feet or 1.524 m which can be seen in the

table below.

Table 2. Aisle Width Estimation from Tompkins et al.

Type of Flow	Aisle Width (feet)
Tractors	12
3-ton forklift	11
2-ton forklift	10
1-ton forklift	9
Narrow aisle truck	6
Manual platform truck	5
Personnel	3
Personnel with doors opening into the aisle from one side	6
Personnel with doors opening into the aisle from two side	8

After that space requirement can be calculated by using the formula from Sitorus, Rudianto, & Ginting (2020) theory. The pallet used by PT Sari Warna Asli V Textile Industry has a capacity of 2 bales of thread, so the pallet requirement for each type of thread can be

found by dividing the number or amount of inventory of each type of thread by two.

$$S_j = \frac{\text{Maximum storage}}{\text{Capacity of goods per block}} \dots\dots\dots (2)$$

$$S_j \text{ Thread BL} = \frac{7,75}{2} = 3,875 \approx 4 \text{ pallet}$$

Table 3. Space Requirement of Each Type of Goods

Thread Type	Average Daily Inventory (bale)	Space Requirement (pallet)
GRADE BL	7,75	4
12S PE WV	6	3
20S PE WV	29,97	15
20S PE WAX	254	128
24S PE WAX	263,59	132
30S PE WV	64,01	33
30S PE WAX	225,08	113
40S PE WV	11,66	6
Total	862,06	434

Then calculate the throughput value based on the number of removed goods divided by the number of goods in one transport from the storage location to the exit. The hand lift used by PT Sari Warna Asli V Textile Industry can transport 2 bales of thread. Sitorus, Rudianto, & Ginting (2020) formulate the throughput formula as follows:

$$T_j = \frac{\text{Amount goods accepted - removed}}{\text{Amount of goods removed per transport}} \dots (2)$$

Since the focus of this research is to determine the layout that results in the shortest spent distance, the throughput formula used is:

$$T_j = \frac{\text{Amount goods removed}}{\text{Amount of goods removed per transport}} (3)$$

$$T_j \text{ of Grade BL in February} = \frac{0}{2} = 0$$

Table 4. Total Daily Thread Removed Each Month

Thread Type	February	March	April
GRADE BL	0	0	0
12S PE WV	0	0	0
20S PE WV	340,66	82,3	0
20S PE WAX	471	386,25	137
24S PE WAX	1014	1015	100
30S PE WV	219,8	633,43	800
30S PE WAX	1261	1492	1239,28
40S PE WV	43,975	95	42,46

Table 5. Total Thread Removed Throughput Each Month

Thread Type	Month			Total
	February	March	April	
GRADE BL	0	0	0	0
12S PE WV	0	0	0	0
20S PE WV	171	42	0	213
20S PE WAX	236	194	69	499
24S PE WAX	507	508	50	1065
30S PE WV	110	317	400	827
30S PE WAX	631	746	620	1997
40S PE WV	22	48	22	92

4.2 Dedicated Storage Method Analysis

- a. Sort throughput values from largest to smallest

Table 6. Thread Throughput Ranking

Thread Type	Throughput	Ranking
30S PE WAX	1997	1
24S PE WAX	1065	2
30S PE WV	827	3
20S PE WAX	499	4
20S PE WV	213	5
40S PE WV	92	6
GRADE BL	0	7
12S PE WV	0	8

- b. Determining the number of pallet stacks
After the ranking of each item is known, the next step is to determine the number of pallet arrangements. The pallet arrangement consists of the number of columns and rows of pallets arranged front to back and left to right in the warehouse. Then the number of rows of pallets

is calculated based on the length of the warehouse. And the width of the warehouse is used to determine the number of pallet columns. The finished goods warehouse is 20 m long and 30 m wide with a 1.5 m aisle. To facilitate the movement of goods and to avoid damage to goods due to water seeping through the walls, 3 aisles are needed in the middle, right and left, and 2 aisles in the front and back.

$$\text{Amount of columns} = \frac{\text{Warehouse length} - \text{aisle}}{\text{Pallet length}} \quad (4)$$

$$= \frac{20 \text{ m} - (1,5 \text{ m} \times 3)}{1,1 \text{ m}}$$

$$= 14 \text{ columns of pallet}$$

$$\text{Amount of Grade BL row} = \frac{S_j}{\text{Amount of columns}} \quad (5)$$

$$= \frac{1}{14} = 0,071$$

$$\approx 1 \text{ row of pallet}$$

Table 7. Number of Pallet Stacks

Thread Type	Space Requirement (Sj)	Jumlah Baris	Jumlah Kolom
30S PE WAX	113	9	14
24S PE WAX	132	10	14
30S PE WV	33	3	14
20S PE WAX	128	10	14
20S PE WV	15	2	14
40S PE WV	6	1	14
GRADE BL	4	1	14
12S PE WV	3	1	14
Total	434		

4.3 Analysis Using the Classified Storage Method

According to the Pareto law explained by Muntasir (2019) for types of goods other than medicine, the class classification in the class-

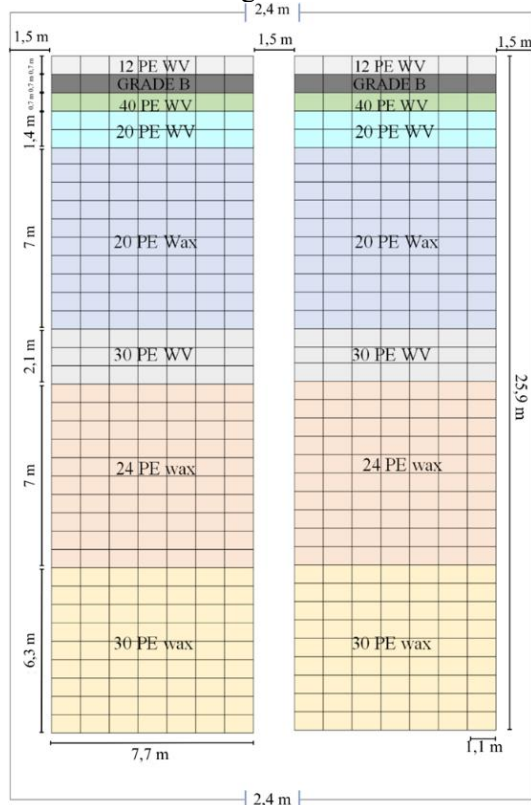
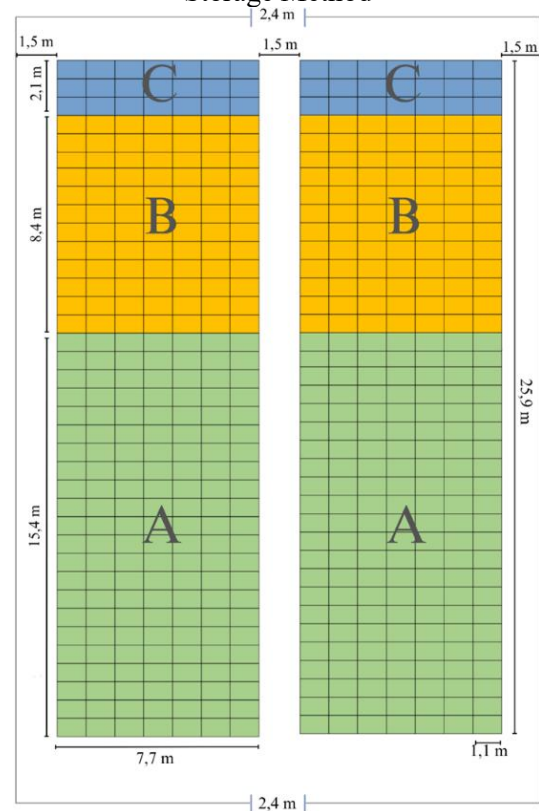
based storage method is class A, B, and C, which contain types of thread with a value of 80%, 15%, and 5% of the investment value or in this case the throughput value, respectively.

Table 8. Classifier for Each Thread Type

Thread Type	Throughput	Presentation	%Cumulative	Class
30S PE WAX	1997	43%	43%	A
24S PE WAX	1065	23%	65%	
30S PE WV	827	18%	83%	
20S PE WAX	499	11%	94%	B
20S PE WV	213	5%	98%	
40S PE WV	92	2%	100%	C
GRADE BL	0	0%	100%	
12S PE WV	0	0%	100%	
Total	4693			

4.4 Overview of Proposed Layouts

After obtaining the throughput value of each type of thread, a proposed layout is formed that can be applied in the finished goods warehouse.

Figure 1. Proposed Layout Using Dedicated Storage Method**Figure 2.** Proposed Layout Using Class-Based Storage Method

The next step is to determine the distance traveled on both proposed layouts. The material handling distance can be calculated by measuring the distance perpendicularly from the point of entry to the point of exit, or by using the straight-line formula. The calculation begins by determining the coordinate points where each type of product is stored. The coordinate point of the exit is assumed to be at point 0,0, since the exit is the beginning and end of the goods issue activity. The formula used to calculate the distance is:

$$d_{ij} = |x_i - x_j| + |y_i - y_j|$$

Description:

d_{ij} = Distance facility i center facility j

x_i = x coordinate of center of facility i

x_j = x coordinate of center of facility j

y_i = y coordinate of center of facility i

y_j = y coordinate of center of facility j

Table 9. Distance of Each Thread Type in Layout with Dedicated Storage Method

Thread Type	x_j (dm)	y_j (dm)
30S PE WAX	4,8	5,7
	-4,8	5,7
24S PE WAX	4,8	12,4
	-4,8	12,4
30S PE WV	4,8	16,9
	-4,8	16,9
20S PE WAX	4,8	21,3
	-4,8	21,3
20S PE WV	4,8	25,5
	-4,8	25,5
40S PE WV	4,8	26,6
	-4,8	26,6
GRADE BL	4,8	27,2
	-4,8	27,2
12S PE WV	4,8	28
	-4,8	28

The following is an example of calculating the material handling distance on a layout with a dedicated storage method for 30S PE WAX thread types calculated by the rectilinear formula:

$$\begin{aligned}
 30S \text{ PE WAX} &= |x_i - x_j| + |y_i - y_j| \\
 &= |4,8 - 0| + |5,7 - 0| \\
 &= 4,8 + 5,7 \\
 &= 10,5 \text{ dm} \approx 1,05 \text{ m}
 \end{aligned}$$

Table 10. Distance Traveled on Layout with Dedicated Storage Method

Thread Type	xj	yj	Distance Traveled (m)	
30S PE WAX	4,8	5,7	1,05	2,1
	-4,8	5,7	1,05	
24S PE WAX	4,8	12,4	1,72	3,44
	-4,8	12,4	1,72	
30S PE WV	4,8	16,9	2,17	4,34
	-4,8	16,9	2,17	
20S PE WAX	4,8	21,3	2,61	5,22
	-4,8	21,3	2,61	
20S PE WV	4,8	25,5	3,03	6,06
	-4,8	25,5	3,03	
40S PE WV	4,8	26,6	3,14	6,28
	-4,8	26,6	3,14	
GRADE BL	4,8	27,2	3,2	6,4
	-4,8	27,2	3,2	
12S PE WV	4,8	28	3,28	6,56
	-4,8	28	3,28	
Total			40,4	

This shows that the total material handling distance required in the layout with the dedicated storage method is 40,4 m.

Table 11. Distance of Each Thread Type in Layout with Class-Based Storage Method

Class	xj	yj
A	4,8	10,3
	-4,8	10,3
B	4,8	22,2
	-4,8	22,2
C	4,8	27,3
	-4,8	27,3

The following is an example of calculating the material handling distance on a layout with a class-based storage method for 30S PE WAX thread types calculated by the rectilinear formula:

$$\begin{aligned}
 30S \text{ PE WAX} &= |x_i - x_j| + |y_i - y_j| \\
 &= |4,8 - 0| + |10,3 - 0| \\
 &= 4,8 + 10,3 \\
 &= 15,1 \text{ dm} \approx 1,51 \text{ m}
 \end{aligned}$$

Table 12. Distance Traveled on Layout with Class-Based Storage Method

Class	xj	yj	Distance Traveled (m)	
A	4,8	10,3	1,51	3,02
	-4,8	10,3	1,51	
B	4,8	22,2	2,7	5,4
	-4,8	22,2	2,7	
C	4,8	27,3	3,21	6,42
	-4,8	27,3	3,21	
Total			14,84	

This shows that the total material handling distance required in the layout with the class-based storage method is 14.84 m.

4.5 Comparison of the Current Layout and the Two Proposed Layouts

The next step is to analyze and compare the initial layout with the proposed layout. This activity is carried out with the aim of knowing the most suitable layout to be applied at PT Sari Warna Asli V Textile Industry.

Figure 1 shows that the initial layout is irregular. The layout is also not a fixed layout that is used because the placement of goods changes every day. This makes it difficult for officers to find the items they need. In addition, goods are sometimes stored not only in the warehouse but also in other rooms. The scattered storage of goods can hinder the process of releasing goods and the movement of goods. Based on the material handling distance calculation done on the initial layout, it can be seen that the layout produces a fairly high material handling distance. Even these calculations do not include the material handling distance of goods that are sometimes

stored in other rooms.

Figures 2 and 3 show that the layout resulting from the use of dedicated storage and class-based storage methods is neater and more organized. The regularity of the arrangement of goods will certainly make it easier for officers to find and locate the items they need. The movement of goods will also increase, making it easier to apply the FIFO (First In First Out) method. However, based on the calculation of material handling distance on the layout with the dedicated storage method, the resulting mileage is higher than the layout with the class-based storage method. This is because the Classified layout has only three storage bins. Below is a comparison table of the original layout and the two proposed layouts.

Table 13. Comparison of Initial Layout and Second Proposed Layout

Layout	Distance Traveled (m)
Current Layout	Undetected
Layout Dedicated Storage	40,4
Layout Class-Based Storage	14,84

5. CONCLUSION

On the basis of the analysis carried out, the following conclusions can be drawn:

1. The space required for the finished goods warehouse is 286.44 m² to accommodate 737.3 bales of thread or 372 pallets. The aisle required to facilitate the movement of goods is 1.5 m.
2. 30S PE WAX thread has the highest throughput value, followed by 12S PE WV thread and BL grade has the lowest throughput value.
3. Based on ABC analysis, 30S PE WAX, 24S PE WAX and 30S PE WV threads belong to class A. 20S PE WAX and 20S PE WV threads belong to class B. And threads of 40S PE WV, Grade BL and 12S PE WV belong to class C.
4. The proposed layout using the dedicated storage method results in a travel distance of 40.4 m and using the class-based storage method results in a distance of 14.84 m.
6. It can be concluded that the proposed layout using the class-based storage method results in the lowest travel distance, the goods are neatly arranged, and it makes it easier for the officers to find the goods.

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